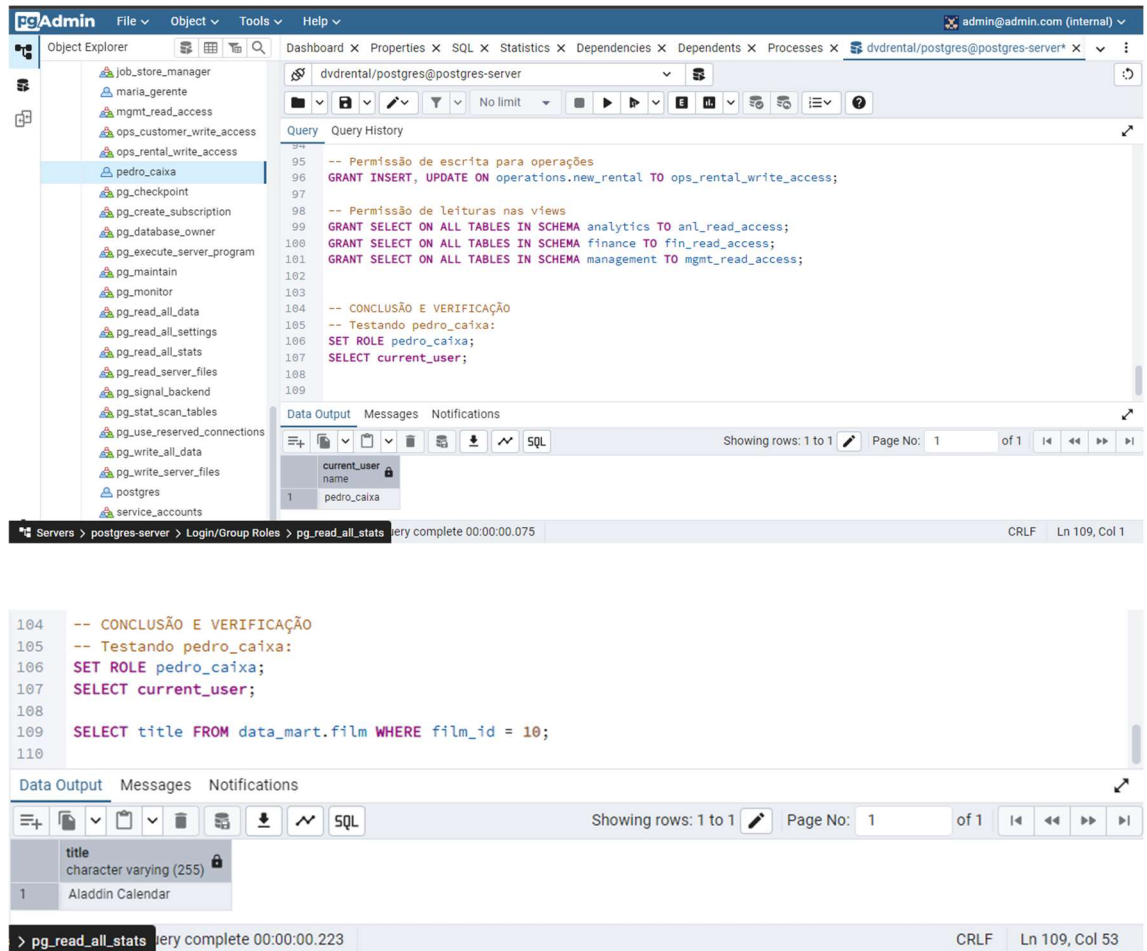


Atividade 5 – Laboratório de Controle de Acesso

Conclusão e Verificação:

1) Testando como usuário Pedro:



The screenshot displays the pgAdmin interface with the following components:

- Object Explorer:** Lists database objects including roles like `pedro_caixa`.
- Query Editor:** Contains SQL queries for granting permissions and testing the user.
- Data Output:** Shows the results of the executed queries.

Query 1 (Lines 95-109):

```
-- Permissão de escrita para operações
GRANT INSERT, UPDATE ON operations.new_rental TO ops_rental_write_access;

-- Permissão de leituras nas views
GRANT SELECT ON ALL TABLES IN SCHEMA analytics TO anl_read_access;
GRANT SELECT ON ALL TABLES IN SCHEMA finance TO fin_read_access;
GRANT SELECT ON ALL TABLES IN SCHEMA management TO mgmt_read_access;

-- CONCLUSÃO E VERIFICAÇÃO
-- Testando pedro_caixa:
SET ROLE pedro_caixa;
SELECT current_user;
```

Data Output 1:

current_user
pedro_caixa

Query 2 (Lines 104-110):

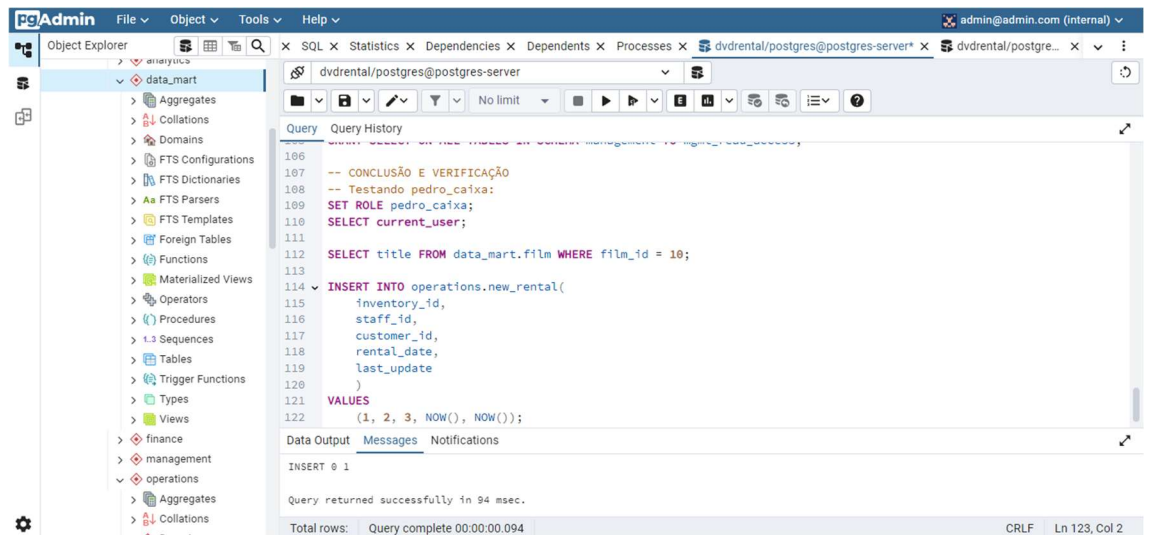
```
-- CONCLUSÃO E VERIFICAÇÃO
-- Testando pedro_caixa:
SET ROLE pedro_caixa;
SELECT current_user;

SELECT title FROM data_mart.film WHERE film_id = 10;
```

Data Output 2:

title
Aladdin Calendar

The status bar at the bottom indicates the current query is `pg_read_all_stats` and is complete.



The screenshot shows the pgAdmin interface with the 'Query' tab selected. The query history displays a successful execution of an SQL script. The script includes comments in Portuguese, setting the role to 'pedro_caixa' and selecting the current user. It then performs an insert into the 'operations.new_rental' table with specific values for inventory_id, staff_id, customer_id, rental_date, and last_update. The status bar at the bottom indicates 'Query returned successfully in 94 msec.' and 'Total rows: Query complete 00:00:00.094'.

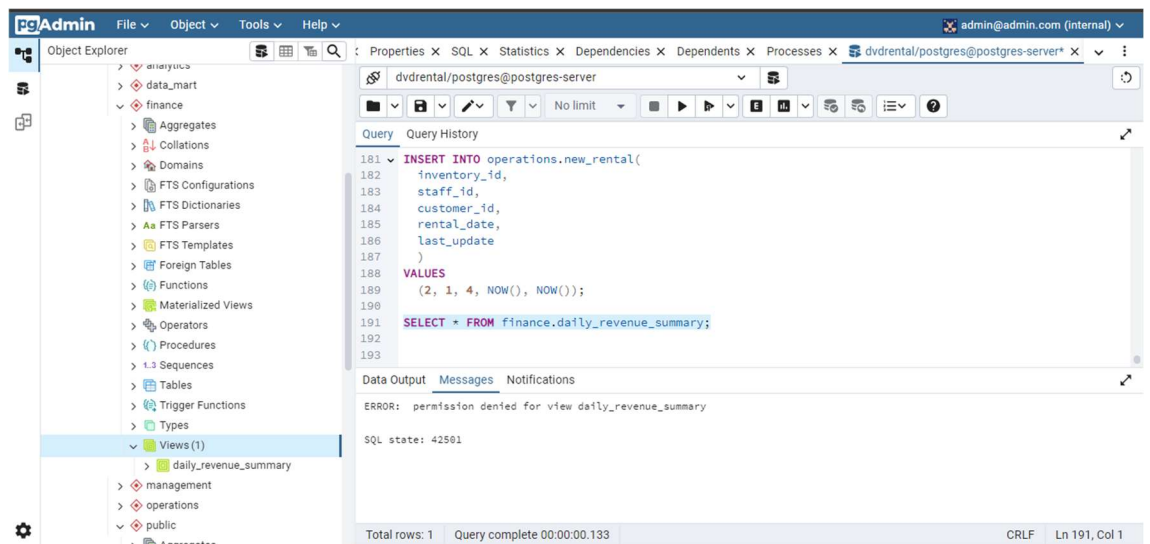
```
106 -- CONCLUSÃO E VERIFICAÇÃO
107 -- Testando pedro_caixa:
108 SET ROLE pedro_caixa;
109 SELECT current_user;
110
111
112 SELECT title FROM data_mart.film WHERE film_id = 10;
113
114 INSERT INTO operations.new_rental(
115     inventory_id,
116     staff_id,
117     customer_id,
118     rental_date,
119     last_update
120 )
121 VALUES
122     (1, 2, 3, NOW(), NOW());
```

Data Output Messages Notifications

INSERT 0 1

Query returned successfully in 94 msec.

Total rows: Query complete 00:00:00.094 CRLF Ln 123, Col 2



The screenshot shows the pgAdmin interface with the 'Query' tab selected. The query history displays an SQL script that attempts to insert data into 'operations.new_rental' and then select from 'finance.daily_revenue_summary'. An error message is shown: 'ERROR: permission denied for view daily_revenue_summary'. The status bar at the bottom indicates 'SQL state: 42501' and 'Total rows: 1 Query complete 00:00:00.133'.

```
181 INSERT INTO operations.new_rental(
182     inventory_id,
183     staff_id,
184     customer_id,
185     rental_date,
186     last_update
187 )
188 VALUES
189     (2, 1, 4, NOW(), NOW());
190
191 SELECT * FROM finance.daily_revenue_summary;
192
193
```

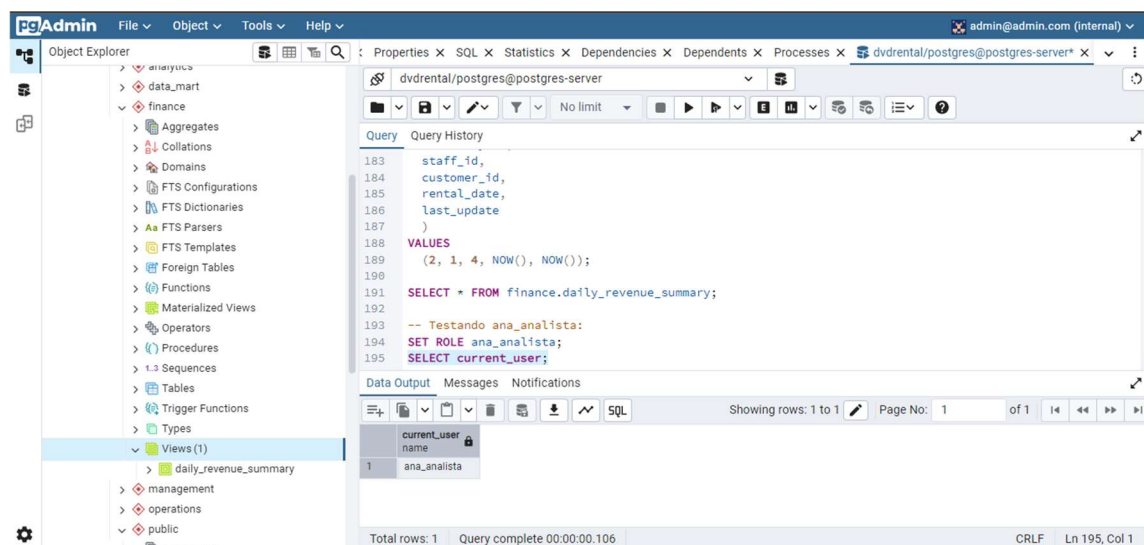
Data Output Messages Notifications

ERROR: permission denied for view daily_revenue_summary

SQL state: 42501

Total rows: 1 Query complete 00:00:00.133 CRLF Ln 191, Col 1

2) Testando como usuário Ana:



The screenshot shows the pgAdmin interface with the 'Query' tab selected. The query history displays an SQL script that sets the role to 'ana_analista' and selects the current user. The status bar at the bottom indicates 'Query complete 00:00:00.106' and 'Total rows: 1'. Below the query history, a table shows the result of the 'SELECT current_user;' query, displaying 'ana_analista'.

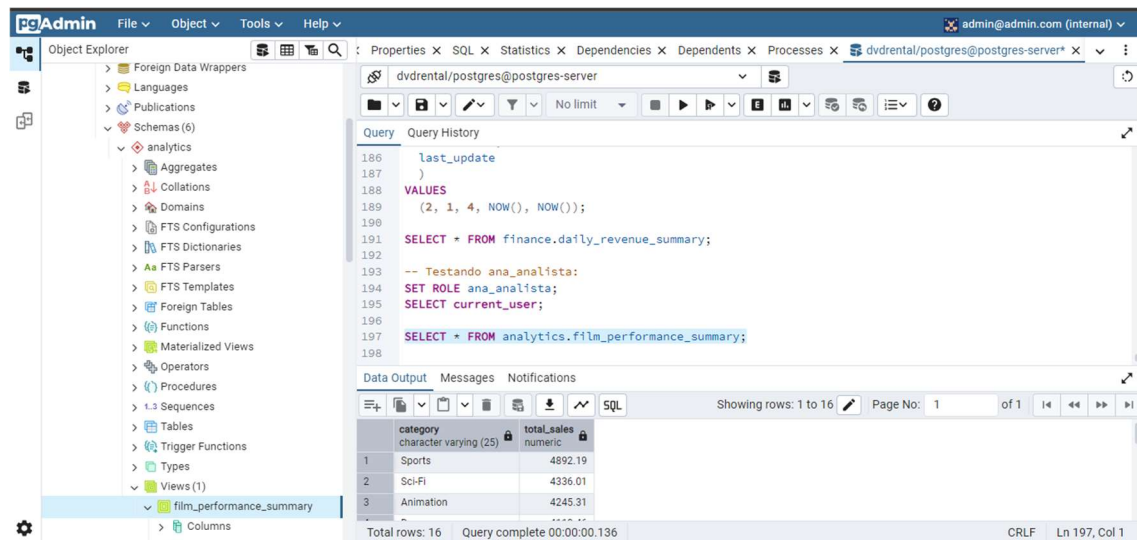
```
183 staff_id,
184 customer_id,
185 rental_date,
186 last_update
187 )
188 VALUES
189     (2, 1, 4, NOW(), NOW());
190
191 SELECT * FROM finance.daily_revenue_summary;
192
193 -- Testando ana_analista:
194 SET ROLE ana_analista;
195 SELECT current_user;
```

Data Output Messages Notifications

current_user
ana_analista

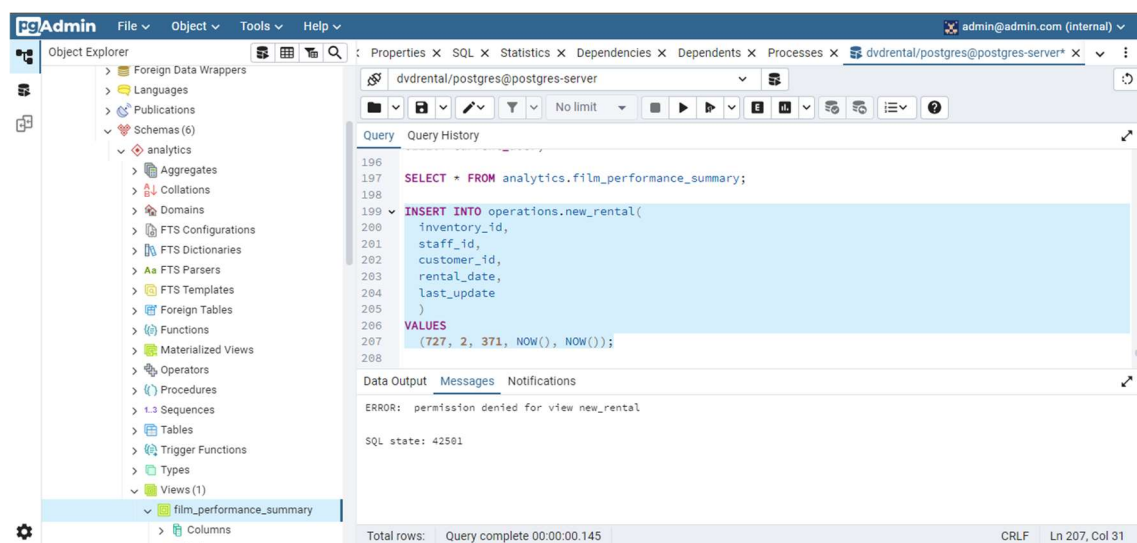
Showing rows: 1 to 1 Page No: 1 of 1

Total rows: 1 Query complete 00:00:00.106 CRLF Ln 195, Col 1



The screenshot shows the PgAdmin interface with a SQL query executed. The query includes a comment about testing a role and a SELECT statement from the `analytics.film_performance_summary` table. The results are displayed in a table with two columns: `category` and `totalSales`.

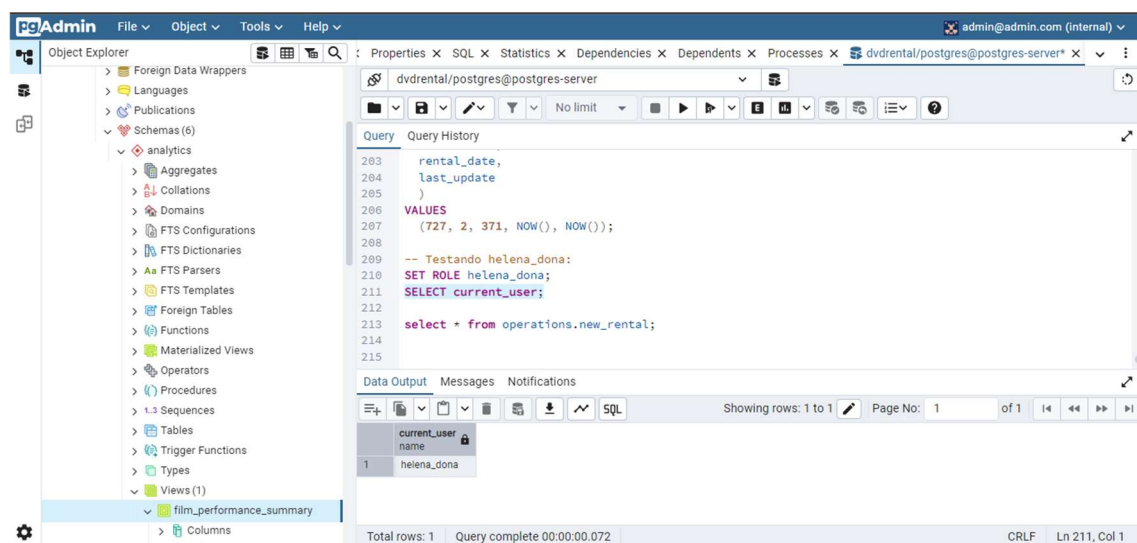
category	totalSales
Sports	4892.19
Sci-Fi	4336.01
Animation	4245.31



The screenshot shows the PgAdmin interface with an SQL error message. The query was an INSERT statement into the `operations.new_rental` table. The error message is: `ERROR: permission denied for view new_rental`.

SQL state: 42501

3) Testando como usuário Helena:



The screenshot shows the PgAdmin interface with a SQL query executed. The query includes a comment about testing a role and a SELECT statement from the `operations.new_rental` table. The results are displayed in a table with one column: `current_user`.

current_user
helena_dona

pgAdmin interface showing a query execution result. The query is:

```
202 customer_id,  
203 rental_date,  
204 last_update  
205 )  
206 VALUES  
207 (727, 2, 371, NOW(), NOW());  
208  
209 -- Testando helena_dona:  
210 SET ROLE helena_dona;  
211 SELECT current_user;  
212  
213 SELECT * FROM analytics.film_performance_summary;  
214
```

The Data Output shows the following table:

category	total_sales
Sports	4892.19
Sci-Fi	4336.01
Animation	4245.31
Prima	4118.46

Total rows: 16 Query complete 00:00:00.106 CRLF Ln 213, Col 1

pgAdmin interface showing a query execution result. The query is:

```
207 (727, 2, 371, NOW(), NOW());  
208  
209 -- Testando helena_dona:  
210 SET ROLE helena_dona;  
211 SELECT current_user;  
212  
213 SELECT * FROM analytics.film_performance_summary;  
214 SELECT * FROM finance.daily_revenue_summary;  
215  
216  
217  
218 SET ROLE postgres;  
219
```

The Data Output shows the following table:

day	total_revenue
2007-05-14	514.18
2007-04-30	5723.89
2007-04-29	2717.60
2007-04-28	2622.73

Total rows: 32 Query complete 00:00:00.179 CRLF Ln 214, Col 1

pgAdmin interface showing a query execution result. The query is:

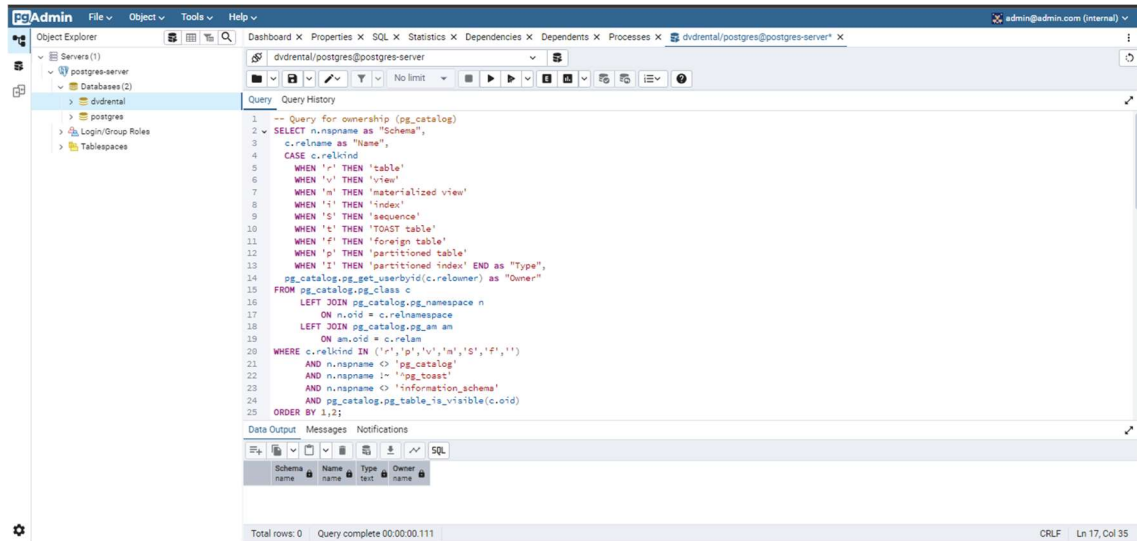
```
205 )  
206 VALUES  
207 (727, 2, 371, NOW(), NOW());  
208  
209 -- Testando helena_dona:  
210 SET ROLE helena_dona;  
211 SELECT current_user;  
212  
213 SELECT * FROM analytics.film_performance_summary;  
214 SELECT * FROM finance.daily_revenue_summary;  
215 SELECT * FROM management.store_performance;  
216  
217  
218  
219
```

The Data Output shows the following table:

store	manager	total_sales
Woodridge,Australia	Jon Steph...	30683.13
Lethbridge,Canada	Mike Hillyer	30628.91

Total rows: 2 Query complete 00:00:00.124 CRLF Ln 215, Col 1

Ownership Query:

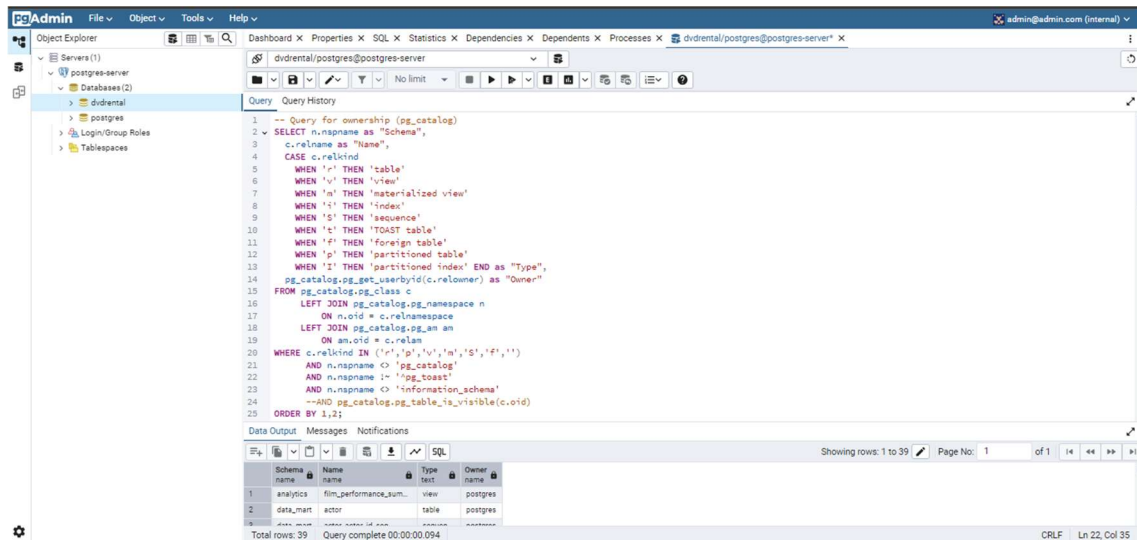


```
1 -- Query for ownership (pg_catalog)
2 SELECT n.nspname as "Schema",
3        c.relname as "Name",
4        CASE c.relkind
5        WHEN 'r' THEN 'table'
6        WHEN 'v' THEN 'view'
7        WHEN 'm' THEN 'materialized view'
8        WHEN 'i' THEN 'index'
9        WHEN 's' THEN 'sequence'
10       WHEN 't' THEN 'TOAST table'
11       WHEN 'f' THEN 'foreign table'
12       WHEN 'p' THEN 'partitioned table'
13       WHEN 'I' THEN 'partitioned index' END as "Type",
14        pg_catalog.pg_get_userbyid(c.relowner) as "Owner"
15 FROM pg_catalog.pg_class c
16 LEFT JOIN pg_catalog.pg_namespace n
17 ON n.oid = c.relnamespace
18 LEFT JOIN pg_catalog.pg_auth_members am
19 ON am.oid = c.relam
20 WHERE c.relkind IN ('r','p','u','m','s','f','I')
21 AND n.nspname <> 'pg_catalog'
22 AND n.nspname <> 'pg_toast'
23 AND n.nspname <> 'information_schema'
24 AND pg_catalog.pg_table_is_visible(c.oid)
25 ORDER BY 1,2;
```

Schema	Name	Type	Owner
--------	------	------	-------

Total rows: 0 Query complete 00:00:00.111

Executando o script ownership acima retornou uma tabela vazia, pois a função `pg_table_is_visible(c.oid)` está filtrando tudo e ela retorna FALSE para objetos que não estão no `search_path` atual. Neste caso, retirei esse filtro, para garantir que se consiga ver tudo que está no banco de dados.



```
1 -- Query for ownership (pg_catalog)
2 SELECT n.nspname as "Schema",
3        c.relname as "Name",
4        CASE c.relkind
5        WHEN 'r' THEN 'table'
6        WHEN 'v' THEN 'view'
7        WHEN 'm' THEN 'materialized view'
8        WHEN 'i' THEN 'index'
9        WHEN 's' THEN 'sequence'
10       WHEN 't' THEN 'TOAST table'
11       WHEN 'f' THEN 'foreign table'
12       WHEN 'p' THEN 'partitioned table'
13       WHEN 'I' THEN 'partitioned index' END as "Type",
14        pg_catalog.pg_get_userbyid(c.relowner) as "Owner"
15 FROM pg_catalog.pg_class c
16 LEFT JOIN pg_catalog.pg_namespace n
17 ON n.oid = c.relnamespace
18 LEFT JOIN pg_catalog.pg_auth_members am
19 ON am.oid = c.relam
20 WHERE c.relkind IN ('r','p','u','m','s','f','I')
21 AND n.nspname <> 'pg_catalog'
22 AND n.nspname <> 'pg_toast'
23 AND n.nspname <> 'information_schema'
24 --AND pg_catalog.pg_table_is_visible(c.oid)
25 ORDER BY 1,2;
```

Schema	Name	Type	Owner
analytics	film_performance_sum...	view	postgres
data_mart	actor	table	postgres

Total rows: 39 Query complete 00:00:00.094